

A.8 GenAI Keywords in Worker Position Descriptions

This appendix reports the full regression results underlying the validation exercise in Section 4.2.3. The main text presents the time-series pattern visually (Figure 3); here we describe the keyword list and formalize the pattern in a difference-in-differences framework.

We apply a curated list of GenAI keywords to 18.3 million U.S. position descriptions spanning 2021 Q1 through 2025 Q1, drawn from the Revelio Labs individual-positions data. The keyword list builds on the list used to identify GenAI integrator postings in Section 4.2, augmented with additional terms that workers are more likely to use when describing their own adoption of GenAI tools on their résumés. Matching is case-insensitive; ambiguous short forms (e.g., LLM, GPT, NLP, Bard) are matched with word-boundary regex to avoid false positives. Positions with descriptions shorter than 20 characters are excluded as likely placeholders.

The full keyword list is as follows:

- **Original list (job-postings analysis).** *Models and products:* ChatGPT, Chat GPT, GPT, GPT-4, GPT-3, GPT-4o, GPT integration, Claude, Copilot, Gemini. *Concepts:* large language model (LLM), generative AI, Gen AI, transformer-based model, prompt engineering, prompt design, vector embeddings, vector database, retrieval-augmented generation (RAG). *Frameworks, APIs, and infrastructure:* LangChain, LlamaIndex, RetrievalQA, Pinecone, Weaviate, Milvus, OpenAI API, Anthropic Claude API, Azure OpenAI, Google Vertex AI Generative, HuggingFace Transformers.
- **Additional résumé-specific terms.** *Worker self-description of AI use:* AI tools, AI-powered, AI-driven, AI-assisted, AI-based, AI automation. *Adjacent ML/DL vocabulary:* machine learning, ML model, deep learning, neural network, natural language processing (NLP), computer vision. *Consumer-facing GenAI tools:* GitHub Copilot, Midjourney, DALL-E, Stable Diffusion, Bard, Perplexity. *Workflow terms:* fine-tuning, text generation, image generation, AI chatbot, conversational AI.

The additional terms are designed to capture how workers themselves describe GenAI use on their résumés, which differs in tone and vocabulary from how firms advertise integrator roles in vacancy postings: workers are more likely to name the specific consumer-

facing tools they use (e.g., Midjourney, GitHub Copilot) and to describe their work in adjacent ML/DL or workflow terms, while job postings emphasize technical infrastructure (vector databases, frameworks, APIs).

Table A.3 reports the difference-in-differences estimates. The dependent variable is an indicator for whether the position description contains at least one GenAI keyword. Column (1) reports the basic OLS estimate; column (2) adds firm and month fixed effects, yielding a DiD coefficient of 1.35 percentage points (s.e. = 0.05, $p < 0.001$); and column (3) additionally controls for description length.

Table A.3: Difference-in-Differences: GenAI Keywords in Position Descriptions

Dependent Variable:	has_ai		
Model:	OLS (1)	FE (2)	FE + Length (3)
Adopter $\times$ Post	0.0176*** (0.0015)	0.0135*** (0.0005)	0.0140*** (0.0005)
<i>Fixed effects</i>			
Firm		Yes	Yes
Month		Yes	Yes
Observations	18,345,320	18,345,320	18,345,320
R ²	0.010	0.117	0.121
Notes: Standard errors clustered by firm in parentheses. *** $p < 0.01$.			

A.9 Aggregate Employment Trends by Seniority

This appendix presents the aggregate time series of average junior and senior employment across sample firms, referenced in Section 4.1.3.

Figure A.17 presents the aggregate time series of average junior and senior employment across firms. Between 2015 and 2022, employment for both groups expanded at a similar pace, aside from a temporary decline in junior employment during the COVID period. Beginning in mid-2022, however, a clear divergence emerges: senior employment continues to expand steadily, while junior employment plateaus and then, by mid-2023, begins to fall. This divergence is consistent with the findings of Brynjolfsson et al. (2025a), who document a comparable pattern in U.S. payroll data by worker age. The consistency between their results and ours provides external validation for the patterns observed in our LinkedIn-based dataset. In line with these findings, Supplemental Appendix A.10 shows that the unemployment rate among recent college graduates has risen even as overall unemployment has remained largely stable.

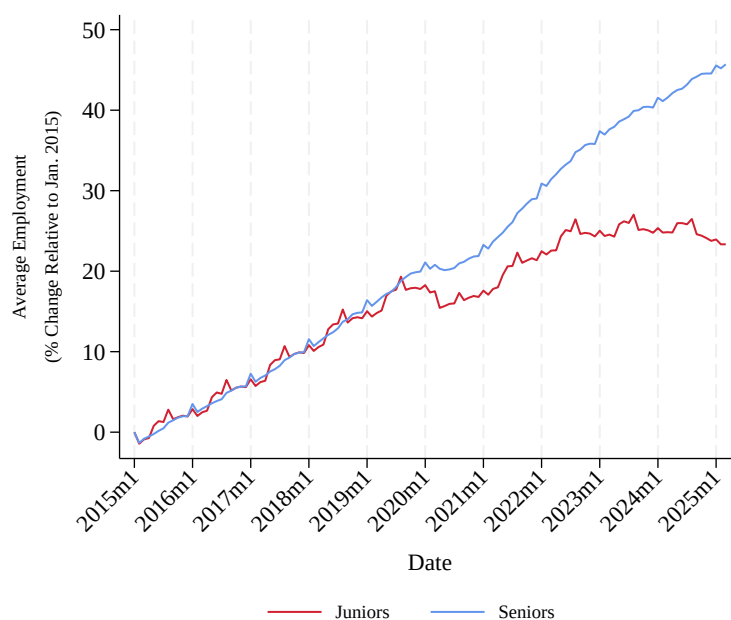


Figure A.17: Log Average Employment of Juniors and Seniors in Sample Firms

Notes: This figure plots the percentage change in the average number of junior- and senior-level workers across firms in our sample over time. Values are normalized to zero in January 2015. “Junior” refers to Entry- and Junior-levels, while “Senior” refers to Associate level and above (see Section 4.1 for details).

A.10 College Graduates Unemployment

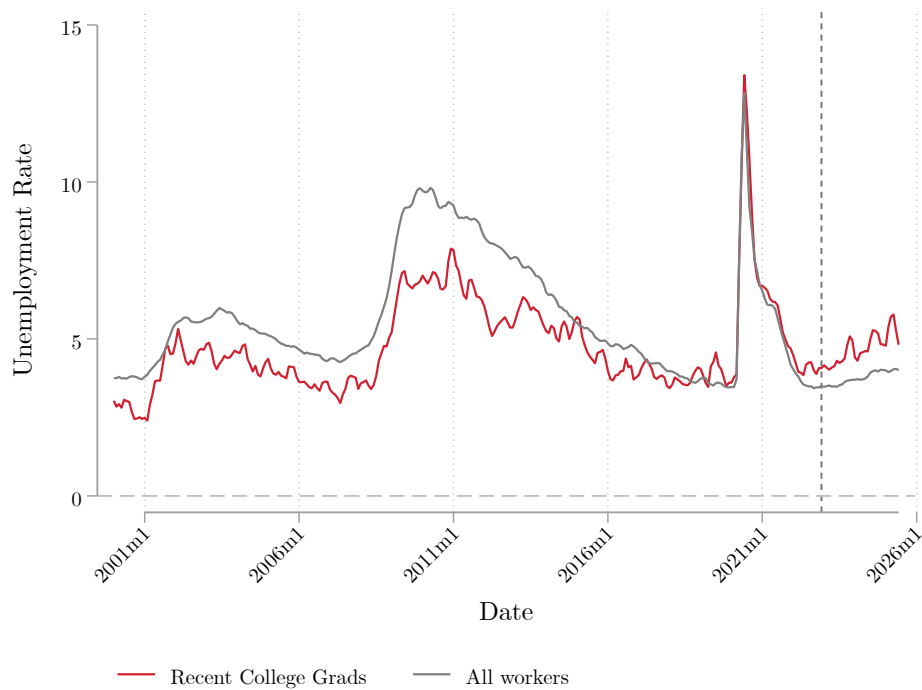


Figure A.18: Unemployment Rates for Recent College Graduates vs. All Workers

Notes: Source: Federal Reserve Bank of New York, *Labor Market for Recent College Graduates*. The red line shows the unemployment rate for recent U.S. college graduates (aged 22–27 with a bachelor’s degree), and the blue line shows the unemployment rate for all U.S. workers, on a monthly basis. Since late 2022, the college-graduate rate has risen even as the overall young worker rate remained mostly flat.